Rasuwa Flood Portal: An Open, Configuration-Driven 3D WebGIS for Rapid Impact Visualisation of the 2026 Bhotekoshi Debris Flood, Nepal

Tek Kshetri, GeoDev

On 26 August 2026, a rock–ice collapse near Langtang Lirung triggered a debris flood that travelled about 72 km down the Bhotekoshi–Trishuli river system in central Nepal, affecting 24 municipalities across eight districts. In the weeks that followed, information critical to response was produced quickly but remained fragmented. Flood extents, damage assessments, landslide inventories and pre- and post-event satellite imagery were spread across government agencies, humanitarian mapping communities, research groups and satellite platforms, in inconsistent formats and with limited public access. This study presents the Rasuwa Flood Portal, an open, browser-based 3D WebGIS that brings these sources together in a single, continuously maintained resource. The portal integrates 40 map layers from more than ten open sources, rendered over terrain with MapLibre GL JS. These include OpenStreetMap-based impact layers, HOT/HDX damage assessments, a British Geological Survey landslide inventory, Vantor post-event imagery and Copernicus Sentinel-2 L2A imagery. A swipe-comparison module contrasts Sentinel-2 True Color, False Color and NDVI composites from before (12 August) and after (21 September) the event. An elevation-profile module relates the progression of the flood surge to the river's longitudinal profile. The integrated layers document approximately 145 km of damaged road and 163 km of endangered road, 190 exposed settlements, 915 mapped landslides and a 45 km² observed flood extent, together with affected schools, health facilities, hydropower projects and banks. The key innovation is a configuration-driven design: every layer, its styling and its popups are declared in a single JSON file. New datasets can therefore be added without programming, and the portal evolves as data are refined. The work shows that lightweight, open-source WebGIS can turn fragmented post-disaster Earth-observation and crowdsourced data into an accessible decision-support tool for responders, researchers and the public. It also offers a transferable template for rapid impact mapping of future cascading hazards in the Himalaya.

Keywords: debris flood; glacial hazard; WebGIS; Sentinel-2; NDVI; open data; disaster risk reduction; Nepal